

Hassan Raza

Software Developer & Data Analytics Professional

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PROFESSIONAL SUMMARY

Software Developer and Data & Analytics Professional with hands-on experience across full-stack web development, database administration, Machine Learning, Business Intelligence, and desktop/mobile application development. Led the end-to-end redesign, security remediation, and live production deployment of a university website serving real students and staff, alongside building analytics dashboards, AI-powered solutions, and production-ready systems using PHP, MySQL, Python, SQL, Streamlit, Flask, and Firebase. Strong ability to independently own a project from architecture through deployment, transforming requirements and data into reliable, real-world systems.

CORE SKILLS

Web Development & Databases: PHP, MySQL, JavaScript, HTML/CSS, Tailwind CSS, Next.js, RESTful integration, database schema design & migration

Data Analysis & Reporting: SQL (PostgreSQL, SQLite, MySQL), Cloud Databases (Google Firebase, Supabase), EDA, KPIs, Dashboards (Streamlit)

Machine Learning & Modeling: Regression, Classification, NLP (TF-IDF), Anomaly Detection, Model Evaluation

Data Applications & Engineering: Python Data Apps, Streamlit Multi-page Systems, Data Modeling, Python (Tkinter), Flask, Desktop Apps

Mobile Application Development: Kotlin, Android Studio

Deployment & Delivery: Live Production Deployment, Desktop Executables (.exe), Cloud Deployment, Offline-first Systems

Visualization: Plotly, Matplotlib, Seaborn

SKILLS

- Strong analytical and data interpretation skills
- Exploratory data analysis
- Excellent communication skills
- Regression analysis
- Ability to manage priorities and cross-functional collaboration
- Advanced modeling
- Python, Streamlit, Pandas, Numpy, Scikit-Learn, Seaborn, Plotly, Matplotlib, SQL, Supabase, Google Firebase, PHP, MySQL
- Statistical analytics

PROJECTS

FLAGSHIP PROJECT

Shaikh Zayed Islamic Centre (SZIC), University of Peshawar — Full Website Redesign & CMS (PHP, MySQL, JavaScript, Tailwind CSS)

- Led end-to-end redesign and live production deployment of the official university website (szic.pk), modernizing the UI/UX while preserving legacy admission, alumni, and library subsystems with zero downtime.
- Designed and built a custom Content Management System enabling non-technical staff to independently manage departments, faculty, programs, news, notices, and admissions.
- Conducted a full security audit and identified a 29,000+ record spam injection vulnerability in the alumni registration system; remediated it by implementing CAPTCHA verification and server-side input validation.
- Migrated and extended the production MySQL schema (20+ new columns, 6 new tables) via a live database migration with zero data loss to existing student and admission records.
- Built a multi-step online admission system with real-time client-side validation, automated merit percentage calculation, and automated applicant email confirmation.
- Diagnosed and resolved cross-environment deployment issues including PHP version compatibility, database corruption

recovery, and server configuration debugging on live infrastructure.

Guluna's Beauty Salon — Booking & E-Commerce Platform (Next.js, Firebase, Stripe, Vercel)

- Independently designed and built a full-stack booking and e-commerce platform for a UK-based beauty salon client, from architecture through production launch.
- Implemented secure card-on-file and pay-now payment flows using Stripe, including migration to Stripe live mode for real transactions.
- Built an admin dashboard for appointment management, service configuration, and business operations.
- Integrated Firebase for authentication and real-time data, and Resend for transactional email verification.
- Managed end-to-end domain migration (Wix to Namecheap to Cloudflare) and continuous deployment via Vercel.

Face Recognition Attendance System for University of Peshawar as a Final Year Project FYP (Python, Flask, OpenCV, MySQL)

- Developed a face recognition-based attendance management system for Shaikh Zayed Islamic Centre classroom environments.
- Implemented real-time facial recognition using OpenCV for automated student attendance marking.
- Designed role-based modules for administrators, teachers, subjects, timetable management, and attendance reporting.
- Integrated MySQL database for secure storage of students, faculty, attendance records, and schedules.
- Generated attendance analytics and reports to improve monitoring and academic record management.

GRE AI Coach Mobile Application (Android, Kotlin, Firebase Authentication)

- Designed and published an Android application to help GRE students practice Quantitative, Verbal, and Writing sections.
- Implemented Firebase Authentication for secure user registration and login.
- Developed quiz management, progress tracking, score analytics, onboarding screens, and AI-powered learning features.
- Created modern Material Design UI with offline-first learning support and personalized study experience.
- Managed complete application lifecycle including development, testing, Android App Bundle generation, and Google Play Console deployment process.

Khattak Tax Solutions Management System (Python, CustomTkinter, Pandas, OpenPyXL, XlsxWriter, Matplotlib)

- Developed a desktop-based client and taxation management system for maintaining client records, tax services, payment history, and outstanding balances.
- Implemented secure client registration, searchable client database, payment ledger, and service tracking modules.
- Integrated Excel reporting using OpenPyXL and XlsxWriter for automated report generation and data export.
- Created analytical charts and business summaries using Matplotlib for monitoring collections and client activity.
- Designed an intuitive CustomTkinter interface focused on ease of use for non-technical business staff.
- Deployed as a standalone executable (.exe) for offline business operations.

Perfume Store Analytics Dashboard (Python, Streamlit, PostgreSQL, Supabase)

- Designed and developed an end-to-end analytics dashboard for a perfume retail business.
- Implemented PostgreSQL database hosted on Supabase for secure cloud data storage.
- Built interactive Streamlit dashboards for sales trends, product performance, revenue analysis, and customer insights.
- Enabled real-time data updates and deployment-ready architecture for business decision support.
- Deployed dashboard online with role-ready structure suitable for remote analytics use.

SMC Clinic Management Software (Python, SQLite, CustomTkinter)

- Developed a professional desktop-based Clinic Management System using Python and CustomTkinter.
- The software includes secure login, patient registration, visit history tracking, medicine template management, and a configurable dashboard with real-time statistics.
- Implemented SQLite for structured medical data storage and designed the system for offline clinic usage with future scalability.
- Deployed as an executable (.exe) for offline use by doctor.

E-Commerce Website (HTML, CSS, JavaScript, Firebase)

- Designed and developed a complete e-commerce website for Lawari Food brand.
- Implemented modern, responsive UI using HTML, CSS, and JavaScript.
- Integrated Firebase as backend for database, authentication, and product management.
- Developed admin panel features for product add/update, pricing control, and inventory handling.
- Enabled order management, customer data storage, and real-time updates via Firebase cloud database.
- Optimized website structure for scalability, performance, and future payment gateway integration.

SGCA School Management Software (Python, SQLite, Streamlit)

- Developed a desktop ERP system for a primary school with modules for student management, fee tracking, certificates, and SMS center.
- Deployed as an executable (.exe) for offline use by administrators.

Fake News Detection ML Project (Python, Scikit-learn)

- Implemented ML models (Logistic Regression, Random Forest, SVM TF-IDF) to classify news articles.
- Achieved 80% accuracy with explainability reports for performance metrics.

Olympic Story: 120 Years of History (Interactive Dashboard)

- Built a Streamlit dashboard visualizing Olympic data using Python & Plotly.
- Project link: <https://www.kaggle.com/hassanrazakhattak>
- Live Demo: <https://olympicsdashboardbyhassanraza051.streamlit.app/>

Superstore Sales Dashboard (ML Forecasting & Responsiveness)

- Created a responsive dashboard with ML forecasting models (3-12 month horizon).
- Live Demo: <https://superstoredashboard-byhassanraza.streamlit.app/>

Bellabeat Case Study: Smart Device Data Analysis

- Conducted EDA and SQL analysis for Fitbit data to generate marketing insights.
- Kaggle Link: <https://www.kaggle.com/hassanrazakhattak>

Anomaly Detection in Telecom Networks (ML/DL Project)

- Built ML and deep learning models for traffic anomaly detection.
- Kaggle Link: <https://www.kaggle.com/hassanrazakhattak>

INTERNSHIP

Machine Learning Intern - Neuro App (Remote)

- Completed a structured Machine Learning internship focusing on practical implementation of core ML concepts.
- Worked on NumPy and Pandas data handling, data cleaning, feature engineering, and exploratory data analysis.
- Built classification and regression models using Scikit-learn and evaluated them using standard metrics.
- Developed and documented weekly ML tasks with GitHub repositories, screenshots, and README files.
- Gained hands-on experience in data visualization, model persistence, and end-to-end ML workflows.

[Internship Repositories](#)

EDUCATION

BS Computer Science — From University of Peshawar

CERTIFICATES

Google Career Certificate in Advanced Data Analytics

Google Career Certificate in Data Analytics

Project Management by Google

Machine Learning with Python by IBM

Python for Everybody by University of Michigan

Using Databases with Python by University of Michigan

LANGUAGES

English, Urdu, Pashtu